Category: Smart cities

PS-003: Federated Traffic-Flow Prediction with Communication-Cost Evaluation

- **Description:**
  - Centralized traffic forecasting requires edge cameras and roadside sensors to continuously stream raw data to a central cloud, consuming high cellular bandwidth and concentrating private travel logs. Federated Learning addresses this by training local models on edge devices and sharing only mathematical weight updates with a central server. In this project, you will build a federated simulator using a public traffic dataset partitioned across virtual intersections. You will evaluate how well a compact forecasting model using Federated Averaging (FedAvg) handles non-identically distributed (non-IID) traffic patterns, reporting prediction accuracy (MAE and RMSE) against network byte transmission costs per training round.
- **Challenge:**
  - Compare federated and centralised traffic-flow prediction using a public traffic dataset distributed across simulated intersections. Study how non-IID data and communication overhead affect prediction performance and federated learning convergence.
- **Key Requirements**
  - Use one public traffic dataset split across 3–5 simulated clients.
  - Implement FedAvg with one compact forecasting model.
  - Use chronological train/test splits to avoid data leakage.
  - Compare IID and non-IID client distributions.
  - Measure MAE/RMSE, communication bytes per round and convergence.
  - Run experiments using 3 random seeds.
  - No live cameras, city-wide deployment or formal privacy guarantees.
- **Expected Outcome**
  - Federated learning simulator
  - Centralised and federated baselines
  - Traffic-flow forecasting model
  - Evaluation of IID vs non-IID performance
  - Communication-cost and convergence analysis
  - Final evaluation report with results across 3 seeds
- **Deliverables***
  1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)
     - Each team must set up a private repository on GitHub.
     - Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
     - Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)
     - Ensure that your repository is named according to your team's name.

***All deliverables need to be completed for getting particiaption certificate.**

2. Each team is required to upload a PDF to their GitHub repository
    - Please follow the specified structure: Proposed-structure-hackathon.pdf
3. Prototype Demonstration with a Pseudocode Snippet to the mentor

- **References**
    - https://proceedings.mlr.press/v54/mcmahan17a.html